1. Historical Growth Trend Assessment:

How has the EV market grown over the years?

In [6]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

In [7]:
```
#load Excel File
df = pd.read_excel("Electric_Vehicle_Population_Data_Excel_sheet.xlsx")

print("Data Loaded Successfully")
print(df.head())
```

```
Data Loaded Successfully
   VIN (1-10)     County       City State  Postal Code  Model Year   Make  \
0  5YJYGDEE1L       King    Seattle    WA      98122.0        2020  TESLA
1  7SAYGDEE9P  Snohomish    Bothell    WA      98021.0        2023  TESLA
2  5YJSA1E4XK       King    Seattle    WA      98109.0        2019  TESLA
3  5YJSA1E27G       King   Issaquah    WA      98027.0        2016  TESLA
4  5YJYGDEE5M     Kitsap  Suquamish    WA      98392.0        2021  TESLA

     Model           Electric Vehicle Type  \
0  MODEL Y  Battery Electric Vehicle (BEV)
1  MODEL Y  Battery Electric Vehicle (BEV)
2  MODEL S  Battery Electric Vehicle (BEV)
3  MODEL S  Battery Electric Vehicle (BEV)
4  MODEL Y  Battery Electric Vehicle (BEV)

   Clean Alternative Fuel Vehicle (CAFV) Eligibility  Electric Range  \
0           Clean Alternative Fuel Vehicle Eligible             291
1  Eligibility unknown as battery range has not b...               0
2           Clean Alternative Fuel Vehicle Eligible             270
3           Clean Alternative Fuel Vehicle Eligible             210
4  Eligibility unknown as battery range has not b...               0

   Base MSRP  Legislative District  DOL Vehicle ID  \
0          0                  37.0       125701579
1          0                   1.0       244285107
2          0                  36.0       156773144
3          0                   5.0       165103011
4          0                  23.0       205138552

                Vehicle Location  \
0   POINT (-122.30839 47.610365)
1  POINT (-122.179458 47.802589)
2   POINT (-122.34848 47.632405)
3   POINT (-122.03646 47.534065)
4   POINT (-122.55717 47.733415)

                                Electric Utility  2020 Census Tract
0   CITY OF SEATTLE - (WA)|CITY OF TACOMA - (WA)       5.303301e+10
1                        PUGET SOUND ENERGY INC       5.306105e+10
2   CITY OF SEATTLE - (WA)|CITY OF TACOMA - (WA)       5.303301e+10
3  PUGET SOUND ENERGY INC||CITY OF TACOMA - (WA)       5.303303e+10
4                        PUGET SOUND ENERGY INC       5.303594e+10
```

In [8]:
```
print(df.columns)
```

```
Index(['VIN (1-10)', 'County', 'City', 'State', 'Postal Code', 'Model Year',
       'Make', 'Model', 'Electric Vehicle Type',
       'Clean Alternative Fuel Vehicle (CAFV) Eligibility', 'Electric Range',
       'Base MSRP', 'Legislative District', 'DOL Vehicle ID',
       'Vehicle Location', 'Electric Utility', '2020 Census Tract'],
      dtype='object')
```

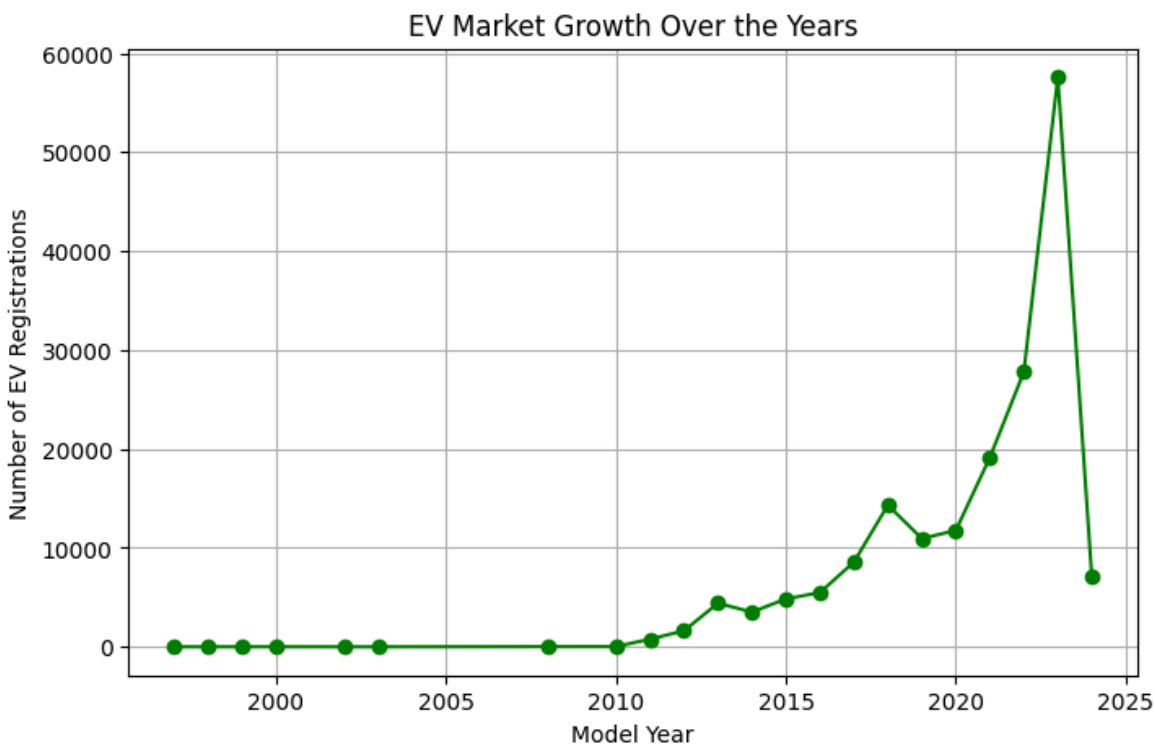
In [9]:
```
# Clean the Model Year Column
df['Model Year'] = pd.to_numeric(df['Model Year'], errors='coerce')
df = df.dropna(subset=['Model Year'])
df['Model Year'] = df['Model Year'].astype(int)
```

In [10]:
```
# Count EVs Per Year
yearly_ev = df['Model Year'].value_counts().sort_index()

print(yearly_ev.head())
```

```
Model Year
1997    1
1998    1
1999    5
2000    7
2002    2
Name: count, dtype: int64
```

In [16]:
```
plt.figure(figsize=(8,5))
plt.plot(yearly_ev.index, yearly_ev.values, marker='o', color='green')
plt.xlabel("Model Year")
plt.ylabel("Number of EV Registrations")
plt.title("EV Market Growth Over the Years")
plt.grid(True)
plt.show()
```



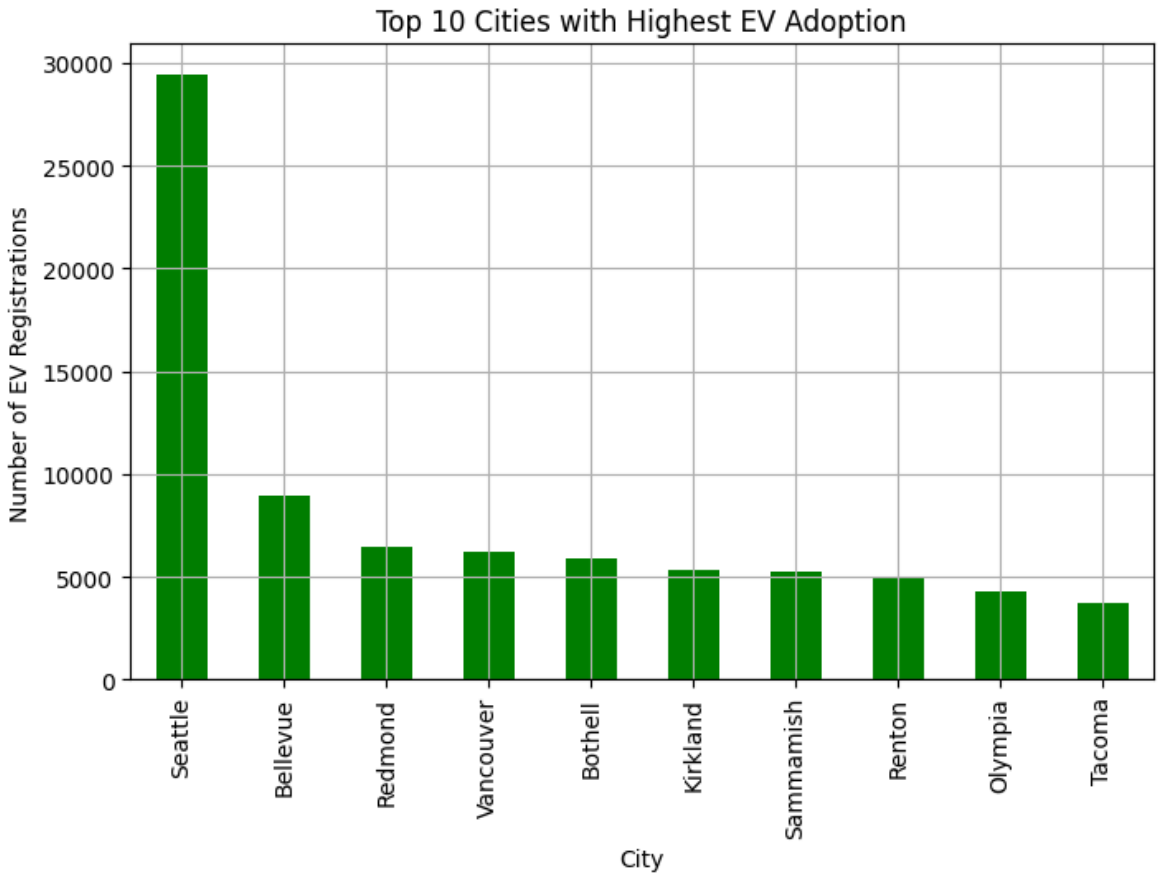
2. Geographical Analysis:

Which regions have the highest EV adoption?

In [20]:
```
top_cities = df['City'].value_counts().head(10)
print(top_cities)
```

```
City
Seattle      29447
Bellevue      8931
Redmond       6478
Vancouver     6193
Bothell       5863
Kirkland      5362
Sammamish     5280
Renton        4980
Olympia       4299
Tacoma        3722
Name: count, dtype: int64
```

In [27]:
```
plt.figure(figsize=(8,5))
top_cities.plot(kind='bar', color='green')
plt.xlabel("City")
plt.ylabel("Number of EV Registrations")
plt.title("Top 10 Cities with Highest EV Adoption")
plt.grid(True)
plt.show()
```



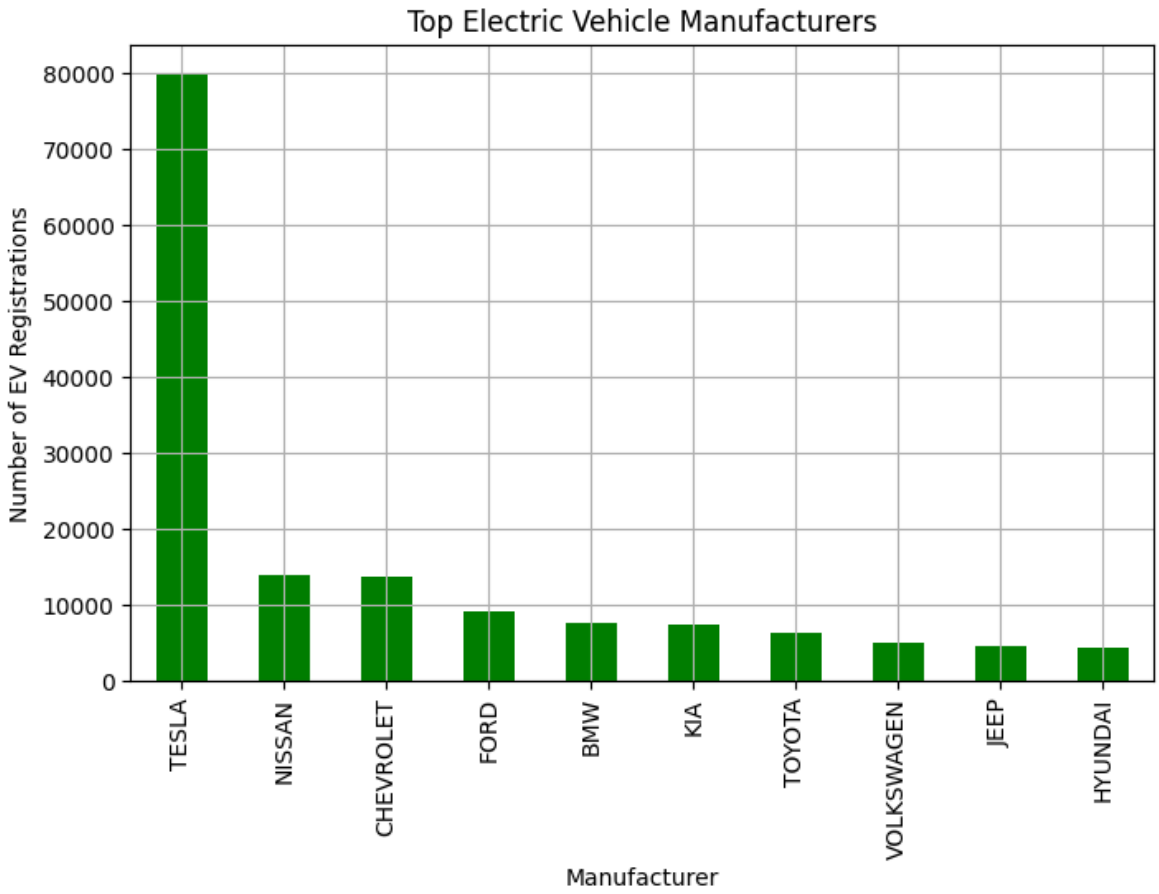
3. Make & Model Analysis:

Which EV manufacturers and models are most popular?

In [22]:
```
top_makes = df['Make'].value_counts().head(10)
print(top_makes)
```

```
Make
TESLA          79659
NISSAN         13998
CHEVROLET      13678
FORD            9199
BMW             7570
KIA             7432
TOYOTA          6288
VOLKSWAGEN      5004
JEEP            4480
HYUNDAI         4406
Name: count, dtype: int64
```

In [25]:
```
plt.figure(figsize=(8,5))
top_makes.plot(kind='bar', color='green')
plt.xlabel("Manufacturer")
plt.ylabel("Number of EV Registrations")
plt.title("Top Electric Vehicle Manufacturers")
plt.grid(True)
plt.show()
```



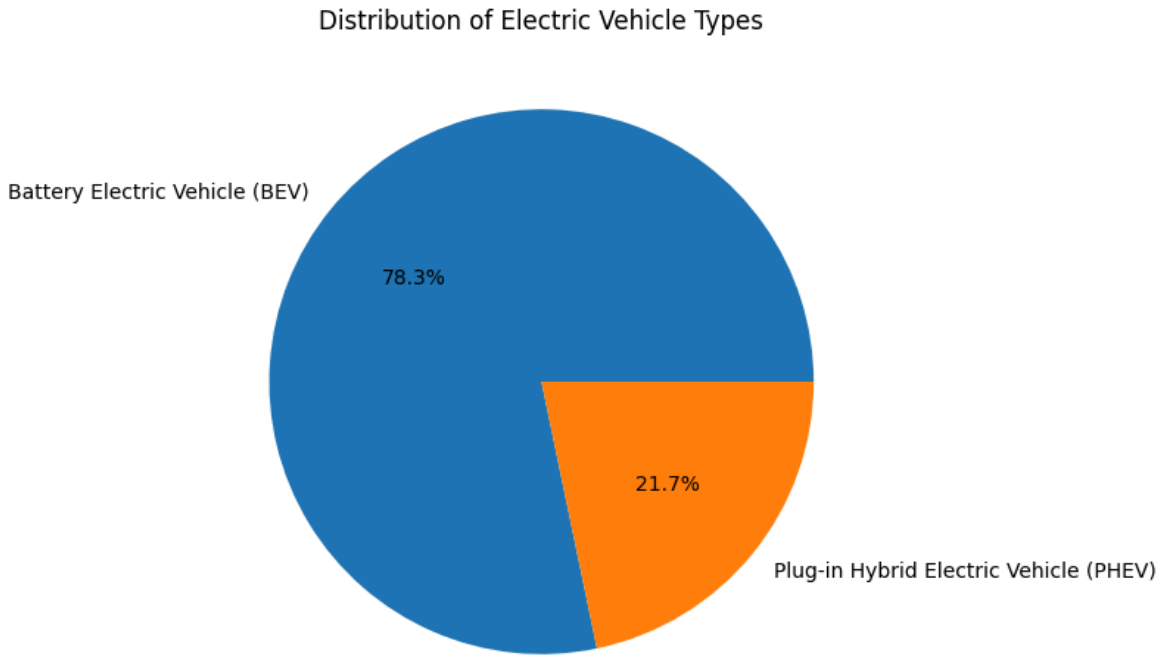
4. EV Type Distribution:

What types of EVs dominate the market?

In [28]:
```
ev_types = df['Electric Vehicle Type'].value_counts()
print(ev_types)
```

```
Electric Vehicle Type
Battery Electric Vehicle (BEV)            139210
Plug-in Hybrid Electric Vehicle (PHEV)     38656
Name: count, dtype: int64
```

In [31]:
```
plt.figure(figsize=(6,6))
ev_types.plot(kind='pie', autopct='%1.1f%%')
plt.title("Distribution of Electric Vehicle Types")
plt.ylabel("")
plt.show()
```



5. Electric Range Analysis:

How has EV technology (electric range) improved?

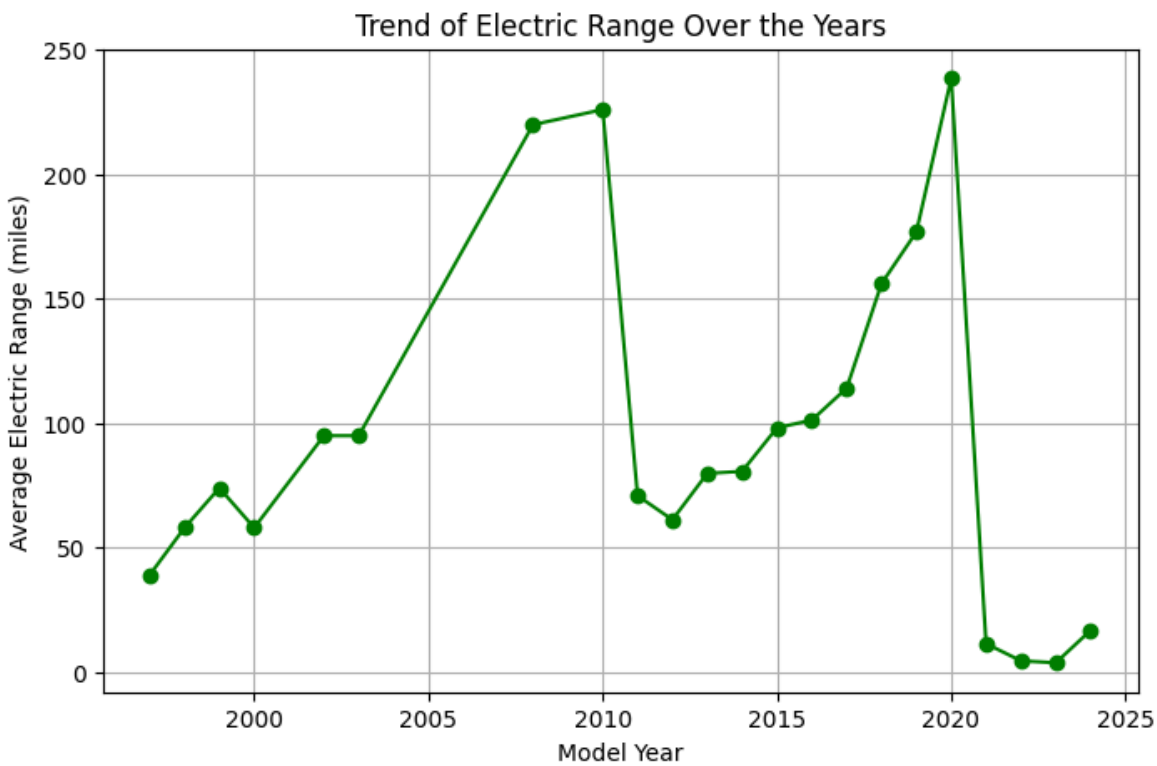
In [32]:
```
df['Model Year'] = pd.to_numeric(df['Model Year'], errors='coerce')
df = df.dropna(subset=['Model Year'])
```

In [33]:
```
avg_range = df.groupby('Model Year')['Electric Range'].mean()
print(avg_range)
```

```
Model Year
1997     39.000000
1998     58.000000
1999     74.000000
2000     58.000000
2002     95.000000
2003     95.000000
2008    220.000000
2010    226.086957
2011     70.891613
2012     61.212608
2013     79.852574
2014     80.669706
2015     98.164740
2016    101.279227
2017    114.124153
2018    156.140194
2019    176.892596
2020    238.690347
2021     11.408426
2022      4.523870
2023      3.731207
2024     16.791667
Name: Electric Range, dtype: float64
```

In [35]:

```
plt.figure(figsize=(8,5))
plt.plot(avg_range.index, avg_range.values, marker='o', color='green')
plt.xlabel("Model Year")
plt.ylabel("Average Electric Range (miles)")
plt.title("Trend of Electric Range Over the Years")
plt.grid(True)
plt.show()
```



6. Market Size Estimation:

What is the current market size of EVs?

In [36]:
```
market_size = len(df)

print("Total Number of Electric Vehicles (Market Size):", market_size)
```

```
Total Number of Electric Vehicles (Market Size): 177866
```

6. Forecasting Using Linear Regression:

What will be the future EV market size?

In [11]:
```
from sklearn.linear_model import LinearRegression

X = yearly_ev.index.values.reshape(-1, 1)
y = yearly_ev.values

model = LinearRegression()
model.fit(X, y)

future_years = [[2025], [2026], [2027]]
predictions = model.predict(future_years)

for year, value in zip([2025, 2026, 2027], predictions):
    print(f"Predicted EV registrations in {year}: {int(value)}")
```

```
Predicted EV registrations in 2025: 20721
Predicted EV registrations in 2026: 21686
Predicted EV registrations in 2027: 22652
```